# AI Engineer Intern Task

Seven-Day Work Log and Development Record

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**Work Log Period**
Sunday, 13 September 2026 – Saturday, 19 September 2026

**Abstract**

This document records the development process followed during the seven-day AI Engineer intern task from Sunday, 13 September 2026, through Saturday, 19 September 2026. The purpose of this log is to document the actual progression of the work rather than only presenting the final result. Each day records the work completed, the technical questions investigated, experiments attempted, problems encountered, corrections made, and the resulting next step. The work evolved from understanding the supplied datasets and schema, through exploratory analysis and multiple segmentation attempts, to independent evaluation, Dataset B workflow segmentation, automation analysis, prototype validation, and final submission preparation.

# Day 1 Sunday, 13 September 2026 – Understanding the Task and Setting Up the Analysis

## 1.1 Main objective

The first day was primarily focused on understanding the problem before implementing a segmentation algorithm. The supplied README, question document, data schema, Dataset A and Dataset B were inspected to understand what the final deliverable was expected to contain.

The main objective for the day was to avoid immediately writing a complicated algorithm without understanding what an "execution", "event", "session", "segment" and "task" meant in the supplied data.

## 1.2 Work completed

The initial work was divided into the following parts:

1. Read the task description and data schema carefully.
2. Inspected the directory structure of Dataset A and Dataset B.
3. Identified the available event logs and metadata.
4. Checked the structure of the JSONL event files.
5. Identified the important event fields, including timestamps, event types, applications, context and text.
6. Started separate analysis scripts rather than modifying the raw data.
7. Established a project structure so that exploration, validation, segmentation and outputs could be kept separate.

The initial project organization was designed around reproducibility. Raw data was kept separate from generated outputs and source code.

## 1.3 Initial observations

Dataset A contained a much larger event stream than Dataset B. The event logs contained a mixture of application switches, screenshots, mouse activity, keyboard activity, browser events, shortcuts and clipboard-related events.

Dataset B was smaller but contained useful textual and application context. This suggested that the two datasets should not necessarily be treated with exactly the same segmentation logic.

A major early observation was that an event stream does not directly contain a clean label saying:

"new task starts here."

Therefore, segmentation had to be inferred from temporal and contextual evidence.

## 1.4 What did not work yet

At this stage there was no reliable segmentation algorithm. It would have been premature to claim that application switches or individual event types were sufficient to identify workflow boundaries.

The decision was therefore made to first quantify the datasets before selecting thresholds.

### 1.5 End-of-day direction

The next step was to build dataset inspection and validation scripts and obtain actual counts rather than making assumptions from a small number of files.

The main principle established on Day 1 was:

**Understand the event stream first; design the segmentation rule second.**

## Day 2 Monday, 14 September 2026 – Dataset A and Dataset B Profiling

### 2.1 Main objective

The second day focused on quantitative dataset analysis. The goal was to determine the scale of the problem, the distribution of event types and applications, and whether the raw data contained enough structure to support deterministic segmentation.

### 2.2 Dataset A inspection

The analysis established:

- 63 sessions,
- 117 event-log files/chunks,
- 162,768 total events,
- approximately 7,300 events containing extracted text,
- 472 events with null active-application information,
- no null context values in the inspected records.

The dominant event types included application switches, keystrokes, screenshots, shortcuts, mouse clicks, browser clicks and clipboard changes.

Application usage was highly concentrated in a few applications, particularly Chrome and Notepad, but the sessions also contained Word, PowerPoint, Explorer, Windows Terminal, Excel and other applications.

### 2.3 Dataset B inspection

Dataset B contained:

- 15 outer session folders,
- 20 nested chunks,
- 20,477 total events,
- 939 events with extracted text,
- 21 event types,

- 11 applications.

The main applications included Edge, Word, Excel, OpenWith, Notepad, Windows Terminal, ProcMine, Explorer and Teams.

The extracted text and window titles showed several recognizable business workflows, including inventory operations, finance-related operations, payroll/HR operations, onboarding and document-based tasks.

### 2.4 Important Dataset B finding

A search was performed for supplier-related evidence because supplier information could have been useful for a later reconciliation prototype.

No supplier-related evidence was found in the 20 Dataset B event files.

This was recorded explicitly rather than inventing supplier information. This later became important for the automation prototype because missing supplier evidence had to result in review rather than an automatic approval.

### 2.5 What did not work

A direct assumption that text alone could identify task boundaries was not supported by the data.

There were many events containing useful text inside the same temporal region, and several different kinds of activity could occur in a single lookback window.

This suggested that:

text evidence alone

was insufficient for reliable start detection.

### 2.6 End-of-day direction

The next step was to investigate temporal structure:

- gaps between events,
- application transitions,
- screenshots around boundaries,
- event density,
- relationship between transitions and ground-truth execution starts.

## Day 3 Tuesday, 15 September 2026 – Boundary Analysis and Early Segmentation Attempts

### 3.1 Main objective

The third day focused on understanding what an actual workflow boundary looked like in Dataset A.

Rather than immediately tuning a segmentation threshold, several possible boundary signals were measured independently.

### 3.2 Application transition analysis

There were 50,588 application-switch events.

However, an important distinction emerged between an event being labelled `app_switch` and an actual change in application identity.

Of the 50,588 application-switch records:

$$7,380$$

represented identity changes, while

$$43,208$$

did not represent a true application identity change.

The actual identity-change rate was approximately:

$$14.59\%.$$

This prevented a simple rule of "every app switch is a new task."

### 3.3 Distance between application transitions and ground-truth starts

Ground-truth execution starts were compared with nearby real application transitions.

The observed results were approximately:

| Distance threshold | Transitions within threshold |
|---|---|
| $\leq 0.5s$ | 1,171 |
| $\leq 1s$ | 1,189 |
| $\leq 2s$ | 1,237 |
| $\leq 5s$ | 1,457 |
| $\leq 10s$ | 3,959 |
| $\leq 20s$ | 5,749 |

This showed that application transitions were useful evidence but did not consistently coincide with task boundaries.

### 3.4 Gap analysis

The raw event gaps were also analyzed.

The approximate distribution included:

$$\text{median}(\Delta t) = 0.057s$$

and

$$\text{mean}(\Delta t) = 0.655s.$$

Longer gaps existed, but they were much less frequent.

This led to an important observation:

**A long gap is evidence of a boundary, not proof of a boundary.**

Using a gap alone could split a long task into multiple pieces.

### 3.5 Screenshot analysis

Screenshots were found to be particularly interesting.

There were 34,580 screenshot events, and 1,905 ground-truth starts had a screenshot as the immediately preceding relevant event.

This suggested that screenshots could act as checkpoint evidence around boundaries.

However, a screenshot itself did not mean that a new task had started. Therefore screenshots were treated as contextual evidence rather than absolute boundaries.

### 3.6 First segmentation direction

The early segmentation experiments therefore combined:

- application transitions,
- browser navigation,
- mouse activity,
- temporal gaps,
- screenshots,
- local activity density.

The initial versions were useful for discovering failure modes but did not yet provide satisfactory precision and recall.

### 3.7 What did not work

The main unsuccessful idea was relying too heavily on a single signal.

In particular:

- application switches over-segmented the stream,
- raw gaps sometimes split the same workflow,
- screenshots were useful checkpoints but were not sufficient by themselves,
- activity events without context created many candidate boundaries.

### 3.8 End-of-day direction

The next step was to create a more structured Dataset A segmentation pipeline and evaluate it against an independently constructed ground truth.

# Day 4 Wednesday, 16 September 2026 – Ground Truth, Independent Evaluation and Iterative Segmentation

## 4.1 Main objective

The fourth day was focused on making the evaluation reliable.

A segmentation algorithm cannot be improved systematically without knowing whether its boundaries are actually close to the ground truth.

## 4.2 Ground-truth construction

The ground-truth analysis established:

- 63 sessions,
- 162,768 raw events,
- 2,009 reconstructed execution records,
- 1,819 explicit `process_started` records.

There were 257 original execution records with missing end timestamps.

Instead of discarding these records, their end boundaries were inferred from the next execution start where possible.

All 257 missing-end cases had a valid next execution start available.

The original and inferred boundary information were kept separately so that the inference was traceable.

## 4.3 Ground-truth alignment

The reconstructed ground truth covered:

$$98.94\%$$

of the raw event stream in the alignment check, with:

$$161,048$$

events inside execution intervals and

$$1,720$$

outside.

This provided a substantially stronger basis for evaluating segmentation.

## 4.4 Independent segmentation evaluation

An independent evaluator was created using one-to-one matching and an IoU threshold:

$$IoU \geq 0.5.$$

Boundary errors were also measured using a 5-second tolerance.

This was important because a segment can have a reasonable overlap even if its start and end are not exactly aligned.

### 4.5 Version comparison

Several Dataset A versions were evaluated.

The progression showed that the segmentation rules were improving, but some versions traded precision for recall.

The important results were:

| Version | Precision | Recall | F1 | Mean IoU | Mean start error |
|---|---|---|---|---|---|
| V4 | 0.1397 | 0.0936 | 0.1121 | 0.6099 | 7.73s |
| V6 | 0.2967 | 0.4719 | 0.3643 | 0.6278 | 5.05s |
| V7 | 0.4647 | 0.5595 | 0.5077 | 0.6637 | 5.94s |
| V8 | 0.4385 | 0.5411 | 0.4844 | 0.6660 | 6.77s |

V7 was selected as the final Dataset A baseline because it provided the strongest overall precision/recall/F1 balance.

V8 had slightly higher mean IoU, but its precision, recall and F1 were lower than V7.

### 4.6 What the experiments taught

The most useful discovery was that workflow boundaries did not have a single universal signature.

Some boundaries were closely associated with:

- screenshots followed by application changes,
- application transitions,
- browser navigation,
- mouse activity.

Other boundaries occurred after longer delays and therefore could not be explained only by an immediate application transition.

### 4.7 End-of-day direction

The Dataset A analysis had now produced a defensible baseline.

The next challenge was Dataset B, where the task required not only identifying the completion of an operation but also identifying where the complete workflow started.

# Day 5 Thursday, 17 September 2026 – Dataset B Segmentation and the Completion-Centred Failure

## 5.1 Main objective

The fifth day focused on Dataset B.

The main challenge was different from Dataset A because Dataset B contained stronger task-specific text and completion patterns.

The first approach was therefore more completion-oriented.

## 5.2 Early Dataset B versions

The earlier versions produced very different segment counts and durations.

The progression included:

| Version | Number of segments | Approx. average duration |
|---|---|---|
| V1 | 188 | 55.4s |
| V2 | 277 | 37.0s |
| V3 | 69 | 150.9s |
| V5 | 151 | 6.06s |
| V6 | 173 | 6.91s |
| V7 | 173 | 4.87s |

These experiments were valuable because they exposed both over-segmentation and under-segmentation.

## 5.3 The major failure

The strict completion-based versions were able to locate reliable completion anchors, but the resulting segments were too short.

The V7 average duration was only:

$$4.87s.$$

This made it clear that finding the final completion event and treating nearby preceding events as the entire workflow was not sufficient.

The core issue was:

> Task completion location does not uniquely determine task start location.

## 5.4 Investigation of possible task-start signals

A dedicated task-start analysis was performed.

For the 173 strong completion anchors:

- the preceding activity included application switches, mouse clicks, mouse scrolling, shortcuts and browser events;

- the 60-second candidate windows often contained several competing activity types;
- the median candidate start distance from completion was approximately 30 seconds;
- all 173 cases had at least a 2-second idle gap somewhere in the lookback;
- 150 of 173 had an idle gap of at least 5 seconds;
- only 41 had an idle gap of at least 8 seconds.

This showed that simply choosing the largest gap would not always identify the correct start.

### 5.5 Important correction

The segmentation problem was reframed from:

Find the completion event

to:

Find the complete activity block leading to a known completion event.

This was the key conceptual change that led to V8.

### 5.6 End-of-day direction

The next version would search backward from the completion anchor, examine activity blocks, use gaps and context changes as evidence, and select an earlier credible start rather than the latest possible event.

## Day 6 Friday, 18 September 2026 – Dataset B V8, Validation and Automation Analysis

### 6.1 Main objective

The sixth day was focused on building the corrected Dataset B V8 pipeline and validating the complete output.

### 6.2 V8 design

The V8 pipeline used:

completion anchor → backward lookback → activity blocks → candidate boundaries → confidence → segment.

The main parameters were:

$$L = 60s,$$

$$\tau_{\text{strong gap}} = 5s,$$

$$\tau_{\text{very strong gap}} = 8s,$$

$$T_{\max} = 120s.$$

The system also required sufficient subsequent activity before accepting a candidate start.

### 6.3 V8 result

The final V8 segmentation produced:

$$173$$

segments.
The duration statistics were:

$$\bar{T} = 32.63s,$$

$$\text{median}(T) = 30.89s,$$

$$T_{\min} = 1.19s, \qquad T_{\max} = 61.57s.$$

The average number of events per segment was approximately:

$$72.03.$$

The start detection methods were:

$$165 \text{ activity-block boundaries}$$

and

$$8 \text{ fallback activity starts.}$$

Confidence distribution:

$$56 \text{ high}, \qquad 109 \text{ medium}, \qquad 8 \text{ low}.$$

There were no overlapping V8 segment pairs.

### 6.4 Validation

Structural validation confirmed:

- 173 segments loaded,
- no missing task labels,
- no invalid intervals,

- no zero/negative-duration segments,
- no segment exceeding the configured maximum,
- all segment timestamps were internally consistent.

The final segments covered approximately:

94.09 minutes

of observed segment time.

### 6.5 Task distribution

The 173 segments were distributed across eight task types:

| Task type | Segments |
|---|---|
| FINANCE_INVOICE | 29 |
| HR_ONBOARDING | 22 |
| LOGISTICS_INVENTORY | 31 |
| LOGISTICS_ORDER | 26 |
| FINANCE_PAYMENT | 21 |
| HR_ATTENDANCE | 23 |
| HR_PAYROLL | 16 |
| FINANCE_BUDGET | 5 |

The three broad areas were:

$$HR = 61, \qquad Finance = 55, \qquad Logistics = 57.$$

### 6.6 Automation opportunity analysis

A heuristic automation score was computed using task frequency, duration, cumulative effort and confidence.

For example, the highest-scoring task types under the defined heuristic included finance invoice processing, HR onboarding, logistics inventory and logistics order.

These scores were treated as prioritization signals rather than predictions.

The observed total segment time was approximately:

94.09 minutes.

A hypothetical 70% savings assumption gave:

65.87 minutes

of potential saving.

This was explicitly treated as a scenario calculation, not as a measured production improvement.

### 6.7 Reconciliation prototype

The automation prototype was also validated.

The extractor could identify invoice identifiers and amounts from available evidence.

However, supplier evidence was not present in the current Dataset B event data.

Therefore, the safe decision was:

$$\boxed{\text{Missing supplier evidence} \Rightarrow \text{REVIEW}}$$

rather than automatic approval.

The current prototype processed 27 invoice observations and kept them in the review path because the evidence required for safe automatic reconciliation was incomplete.

### 6.8 End-of-day direction

By the end of Day 6, the segmentation and prototype logic were functionally complete.

The remaining work was focused on final output consistency, documentation, reproducibility and submission preparation.

## Day 7 Saturday, 19 September 2026 – Final Validation, Packaging and Submission Preparation

### 7.1 Main objective

The final day was dedicated to treating the project as a submission rather than continuing to change the algorithm unnecessarily.

The main goal was to make sure that the final result was:

**correct, reproducible, documented and easy to inspect.**

### 7.2 Final pipeline organization

The final project was organized into separate areas for:

- Dataset A analysis and validation,
- Dataset B analysis,
- segmentation,
- pipeline execution,
- outputs,
- prototype automation,
- reports and documentation.

Historical scripts and experimental outputs were retained for traceability rather than deleting them simply because a later version was selected.

The final Dataset A segmentation was V7.

The final Dataset B segmentation was V8.

### 7.3 Final output schema

The final root-level `segments.jsonl` was checked to contain exactly 173 records.

Each record follows the required compact structure:

```
{
  "session_id": "...",
  "start": "...",
  "end": "...",
  "label": "..."
}
```

The internal richer Dataset B V8 output was retained separately so that confidence and start-detection information was not lost.

### 7.4 Reproducibility checks

The final pipeline commands were organized so that the main analyses could be rerun in sequence.

The final sequence covered:

1. Dataset A analysis,
2. ground-truth construction,
3. ground-truth validation,
4. Dataset A V7 segmentation,
5. independent Dataset A evaluation,
6. Dataset B task-start analysis,
7. Dataset B V8 segmentation,
8. Dataset B validation,
9. segment-quality validation,
10. automation ROI analysis,
11. automation scoring,
12. coverage/interruption validation,
13. final V8 output generation,
14. prototype validation,
15. prototype execution,
16. unit tests.

### 7.5 Git and submission preparation

The final project was also prepared for version control.

A `.gitignore` file was created to prevent raw datasets and very large generated files from being accidentally committed.

The raw data directory was excluded from Git, while source code, documentation, important final outputs and reproducibility files were retained.

The staged project was checked for:

- required README and schema files,
- final `segments.jsonl`,
- final Dataset B outputs,
- source scripts,
- prototype code,
- absence of the large raw dataset,
- absence of the very large generated Dataset A labelled-event file.

This was an important final engineering check because the local project was several gigabytes in size, while the submission repository should contain only the material required to understand, reproduce and evaluate the work.

### 7.6 Documentation

The final documentation was written to distinguish clearly between:

- measured results,
- heuristic assumptions,
- failed experiments,
- final selected versions,
- future improvements.

In particular, the final solution was not described as a neural network.

It was documented as an interpretable temporal segmentation system based on event timing, activity, application transitions, screenshots, task evidence and deterministic boundary logic.

### 7.7 Final technical reflection

The seven-day process showed that the main challenge was not simply writing code.

The more important challenge was deciding which signals actually corresponded to workflow boundaries.

The progression was:

raw logs → dataset understanding → signal analysis → failed assumptions → iterative segmentation → independ

The most important correction was learning that a completion event does not automatically define the beginning of a workflow.

The second important lesson was that no single signal—application switch, screenshot, gap, text or click—was sufficient by itself.

The final system therefore combines multiple weak and strong signals.

### 7.8 What I would improve next

If additional time and labelled data were available, the next step would not be to immediately replace the current algorithm with a large neural network.

Instead, I would first collect more labelled sessions and formulate the problem as temporal boundary detection.

A future model could estimate:

$$P(\text{boundary at } t)$$

using a temporal encoder such as a small Transformer, TCN or recurrent model.

The existing rule-based signals could be retained as input features and safety constraints.

The longer-term architecture would therefore be:

| Raw events → Feature extraction → Rule signals + temporal model → Boundary probability → Segment gener |
|---|

### 7.9 End-of-week conclusion

The seven-day work was deliberately iterative.

Not every version worked, and several experiments produced results that were worse than expected. Those failures were retained as part of the development record because they explained why the final design changed.

The final submission therefore represents not only the final segmentation output, but also the sequence of measurements and corrections that led to it.

**Final principle: measure first, identify failure modes, make a targeted correction, validate independently, and only then finalize the pipeline.**

## Appendix A – Seven-Day Summary

| Date | Focus | Main work | Outcome |
|---|---|---|---|
| 13 Sep | Task understanding | Read requirements, schema and datasets; organized project and inspection workflow. | Problem definition established. |

| | | | |
|---|---|---|---|
| 14 Sep | Dataset profiling | Quantified Dataset A/B events, applications, text and task evidence. | Main signals and limitations identified. |
| 15 Sep | Boundary analysis | Studied app transitions, screenshots, temporal gaps and early segmentation. | Single-signal rules rejected. |
| 16 Sep | Evaluation | Built ground truth, alignment checks and independent IoU evaluation; compared versions. | Dataset A V7 selected. |
| 17 Sep | Dataset B | Tested completion-centred segmentation and analyzed task-start evidence. | Completion-as-start failure identified. |
| 18 Sep | V8 + prototype | Implemented V8 backward activity-block segmentation; validated outputs and automation logic. | Dataset B V8 finalized. |
| 19 Sep | Submission | Final validation, output schema checks, documentation, Git preparation and reproducibility review. | Submission package prepared. |

## Appendix B – Final Numbers at a Glance

| **Metric** | **Value** |
|---|---|
| Dataset A sessions | 63 |
| Dataset A raw events | 162,768 |
| Dataset A reconstructed executions | 2,009 |
| Dataset A final version | V7 |
| Dataset A precision | 46.47% |
| Dataset A recall | 55.95% |
| Dataset A F1 | 50.77% |
| Dataset A mean IoU | 66.37% |
| Dataset B chunks | 20 |
| Dataset B raw events | 20,477 |
| Dataset B final segments | 173 |
| Dataset B V8 mean duration | 32.63s |
| Dataset B V8 median duration | 30.89s |
| Dataset B high-confidence segments | 56 |
| Dataset B medium-confidence segments | 109 |
| Dataset B low-confidence segments | 8 |
| Dataset B observed segment time | 94.09 min |